

Modeling Supply-Chain Networks by a Multi-Agent System

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Abstract

This article proposes a multi-agent information system (MAIS) approach to model the order fulfillment process (OFP) in supply chain networks (SCNs). An order fulfillment process starts with receiving orders from customers and ends with delivery of the finished goods. As manufacturing practice is shifting toward the out-sourcing paradigm, the OFP is more likely to be executed throughout SCNs. It becomes imperative to integrate the OFP into SCNs to improve the OFP. The proposed multi-agent information system (MAIS) approach is used for modeling the OFP in SCNs, and evaluating OFP performance by applying the proposed strategies. The objective of reengineering the OFP is to achieve agility of the process in terms of efficiency, flexibility, robustness and adaptability. A multi-agent simulation platform, called Swarm, is enhanced and applied for modeling the MAIS, and experiments are conducted to simulate the OFP in SCNs in a multi-agent environment. Based on the Swarm simulation platform, we model the OFP in SCNs, simulate the OFP, and then evaluate the potential OFP improvement strategies to identify useful strategies for improving the OFP. The results shed lights on identifying the main effects of various strategies on OFP performance. The insights of utilizing various strategies in different SCNs help redesign the OFP in SCNs.

1. Introduction

An enterprise is a composition of business entities, and its main purpose is to deliver value-added goods or

services to its customers. Products delivered by a manufacturing enterprise are physical goods while products delivered by a service enterprise are services. Despite this distinction, an enterprise is composed of several core business processes. Each business process calls for the cooperation and coordination of different functional entities within the enterprise or outside the enterprise to achieve the goals of the process. A business process is a structured, measured set of activities designed to produce a specified output for a particular customer or market ([7], pp. 5). The number of major processes is between 10 and 20 in some leading companies, such as IBM, Xerox and British Telecom ([7], p.28-29). In general, an enterprise supported by the three core business processes: (1) the product development process, (2) the order fulfillment process, and (3) the customer service process.

In order to remain agile, business processes of an enterprise should be monitored and modified efficiently and effectively. So that an enterprise can stay competitive under a constantly changing business environment. The efforts denoted as *business process reengineering* (BPR) are aimed at fundamentally rethinking and radically redesign business processes in order to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service, and speed ([9], pp. 32). The BPR success lies in the CEO support, and effective methods for process redesign. Information technology (IT) plays a critical role in the BPR efforts. IT enables the process reengineering by allowing process owners to envision the resulting processes prior to the process implementation. IT itself also erects the infrastructure for business process implementation. Therefore, methods for modeling and

evaluating business processes are necessary for envisioning the resulting process and crucial for deriving useful strategies to improve processes.

This research focuses on reengineering *the order fulfillment process*, one of the core business processes, by using a multi-agent information system (MAIS) approach. The MAIS serves two purposes: (1) to model the OFP in SCNs, and (2) to evaluate OFP performance by applying the proposed strategies. The objective of reengineering the OFP is to achieve *agility* of the process in terms of efficiency, flexibility, robustness and adaptability. A multi-agent simulation platform, called *Swarm*, is enhanced and applied for modeling the MAIS, experiments are conducted to simulate the OFP in SCNs, and various OFP improvement strategies are evaluated in a multi-agent business environment. The following sections are used for illustrating these efforts and results.

2. The Order Fulfillment Process (OFP)

An *order fulfillment process* (OFP) starts with receiving orders from the customers and ends with having the finished goods delivered. The *order fulfillment cycle time* is defined as the period from order receiving to products delivery. The OFP is complex because it is composed of several activities, executed by different functional entities, and is heavily interdependent between tasks, resources and agents involved in the process. This process involves the coordination of diverse activities such as sales commitment, credit checking, manufacturing, logistics, accounts receivable, and relationships with external suppliers for purchasing or shipping, which normally take place in several different business units ([7], 1993). The main activities of the OFP can be summarized as follows:

1. order management which receives orders from customers and commits order requests,
2. manufacturing, which includes production scheduling, material planning, capacity planning and shop floor control, and
3. distribution, which considers the logistics such as inventory, transportation, etc.

The main objectives of the OFP can be generalized into two dimensions [6][8][14]:

1. delivering qualified products to fulfill customer orders at the right time and right place, and
2. achieving *agility* to handle uncertainties from internal or external environments.

Agility can be defined as the ability to thrive in a competitive environment of continually and

unpredictably changing market opportunities ([8], p.8). Agility can be expressed as having four underlying principles: delivering value to the customer, cooperating to enhance competitiveness, organizing to master change and uncertainty, and leveraging the impact of people and information ([8], p.73-75; [4]). In business process redesign, the “agility” can be measured in the following dimensions:

1. *efficiency*, defined as the cycle time reduction of business processes.
2. *flexibility*, defined as the minimization of costs involved in dealing with changes in the process.
3. *robustness*, defined as the strength of the information system to handle uncertainty in the process.
4. *adaptability*, defined as the self-learning ability of the information system such that the performance of the business process can be improved incrementally.

3. Integrating the Supply Chain Network with the OFP

The efforts of reengineering the OFP should be spent on improving the process throughout the whole supply chain network because (1) manufacturing practice is shifting to the out-sourcing paradigm, (2) activities of the OFP are executed inside a SCN, and (3) the variation of the OFP determines the SCN structure. A supply chain network (SCN) is a network of autonomous or semi-autonomous business entities that are involved, through upstream and downstream linkages, in the different processes and activities that produce physical goods or services to customers [1][11][12][18]. A SCN can be viewed from different perspectives:

1. **From the material flow perspective:** A SCN is a *network* which links business entities together through various activities. A business entity can be a factory transforming materials into products (or semi-products), a distributor delivering products to the downstream demanders with or without assembly capacity, a retailer selling products to end users, or a warehouse storing materials. The activities within the network, including procurement, manufacturing, and distribution, aim to facilitate the flow of products to customers.
2. **From the roles of entities perspective:** The tasks of each entity are to facilitate the movement of materials from upstream to downstream and, in turn, deliver products to customers. This refers to the

relative roles of entities. A manufacturer is a supplier to its downstream manufacturers and distributors, and at the same time, this manufacturer is also a demander of materials from its upstream suppliers.

3. **From the organizational boundary perspective:** A business entity is a participant of the network, and it can be represented in different degrees of abstraction according to the *organizational boundary*. This contributes the *multiple layer abstraction* of a SCN as shown in Figure 1. This multiple layer abstraction is substantiated by the organizational boundary. Business processes are executed through the *interactions* among the business entities of a SCN. Business processes, in essence, cross organizational boundaries, cooperate different functional units, and exhibit the interdependencies among entities.
4. **From the network visibility perspective:** Business entities in a SCN are linked by *information* and *material* linkages, or either of them, where the relationship between entities can be identified. The information, e.g., data or control, is used to facilitate the flow of materials. Through the information transfer between entities, an entity can have better understanding about the environment, and in turn, make more feasible decisions. The information transfer between entities also enlarges the *visibility* of the SCN. Due to the various degrees of visibility of a SCN, there exists different degree of network transparency of the network structure.

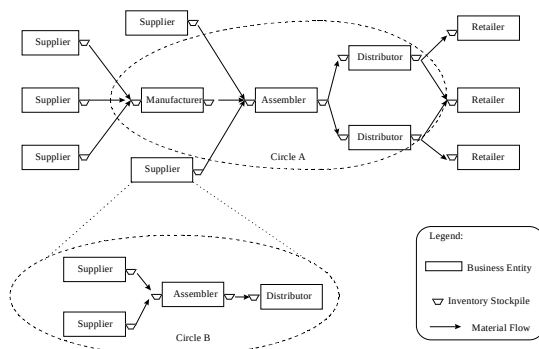


Figure 1. The multiple layer abstraction of a SCN

As described in Section 2, the OFP consists of order management, manufacturing, and distribution activities which are executed within a SCN. Figure 2 displays the order fulfillment process within a supply chain network. The information components, including order management, production scheduling, capacity planning, material planning, and production control, take charge of

the fulfillment of orders. The order information is passed to the order management unit from various downstream business entities (e.g., end users, retailers, or distributors,) depending on the information infrastructure. The information about materials and capacity of upstream business entities is fed in from the first tier supplier or the beyond the first tier suppliers. The functional units in the circle of Figure 2 can be embedded in one business entity (e.g., a factory) for the local manufacturing processes, or they can be implemented to control several entities. The former structure is more distributed planning and control, as the latter one is more centralized planning and control.

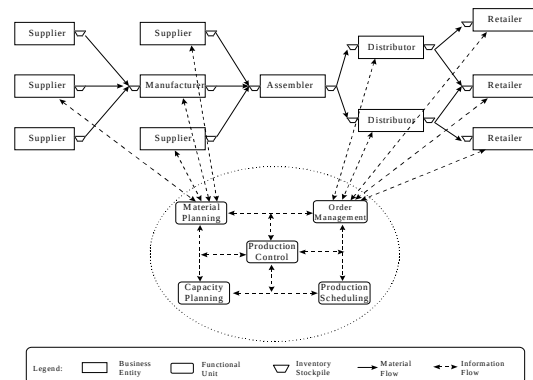


Figure 2. The order fulfillment process within a supply chain network

Issues in improving the OFP within SCNs include (1) how to manage the order information propagation in order to improve the OFP, (2) how to reduce the variability of the OFP within the SCN environment, (3) how to synchronize the availability of materials for manufacturing, (4) how to identify the critical resources and layout the critical path of the network, and put the efforts on the reduction of workload of critical resources, and (5) how to and then change the structure of the network.

4. Multi-Agent Information System Approach (MAIS)

Based on various designs for a multi-agent system in the literature [2][3][5][15][20], and considering the application domain and implementation possibility, we design a multi-agent information system (MAIS) composed of four components: *agents*, *tasks*, *organizations*, and *information infrastructure*. The

components of the MAIS are described as follows:

1. An *agent* is an active object which possesses certain capabilities to perform tasks, and communicates with other agents based on the organizational structure to cooperate the accomplishment of tasks.
2. *Tasks* are things which agents are assigned to work on. Depending on the structure of the agents in an organization, tasks can be *decomposed* by an agent and distributed to other agents, or tasks accomplished by agents are *composed* as the global accomplishment.
3. *Organizations* within a MAIS is formed according to the relations between agents. An organization is formed by the connection through control and information linkages between agents. There are two types of control mechanisms, *centralized* and *decentralized controls*. The control linkage between agent belongs to one of the two types or the hybrid of both mechanisms.
4. *Information infrastructure* lays out a foundation for communications between agents. In MAIS, the digital information storage, operations, and communications technology are used for facilitating the communications and interactions among agents. Information is shared either by common accessible databases or through message passing.

Table 1. The mapping of properties between the MAIS and Swarm

Multi-Agent Information System	Swarm Simulation System
Agent	Object-oriented representation
Self-organization	Nested inherent hierarchy
Autonomous agent behavior	Individual-based modeling
Distributed and loosely coupled control	Parallel and concurrent processing

Agents in the MAIS are distributed geographically, are autonomous but interdependent in their task environments, are embedded with self-organization capability, and are adaptive to changes in their environment. A simulation system should be able to model these features in an efficient and effective way, so that the interaction between agents can be represented, and the system dynamics can be identified. Swarm is a multi-agent simulation software platform for the study of complex adaptive systems. A prototype of the Swarm simulation system was developed at the Santa Fe Institute and aimed to provide a general purpose simulation tool for building simulation models [10][19]. Table 1 describes the mapping of properties between the MAIS and Swarm. The mutli-agent information system

utilizing the Swarm simulation platform has been used for enterprise integration modeling [13], and illustrating electronic virtual organization management [16][17].

5. Modeling the OFP in SCNs

In this research, a MAIS behaves dual roles in the context of reengineering the OFP in SCNs. The MAIS acts as a *modeling tool* to represent the business entities as agents and the involved information and material flows for the OFP in SCNs. The MAIS also acts as a *testbed* for evaluating proposed strategies in improving the OFP in SCNs. The OFP consists of two types of movements: *material flow* and *information flow*. We design two kinds of agents in this environment: *physical agents* and *logical agents*. A physical agent represents a physical existing objects, such as a corporate, factory, production line, machine, etc. A logical agent represents a logical existing object with certain function, such as MRP system, order entry system, production control system, etc. The main distinction between physical and logical agents lies in that a logical agent is more active than a physical agent. The MAIS model for the OFP in a SCN utilizes physical and logical agents, and the corresponding material and information flows are shown in Figure 3.

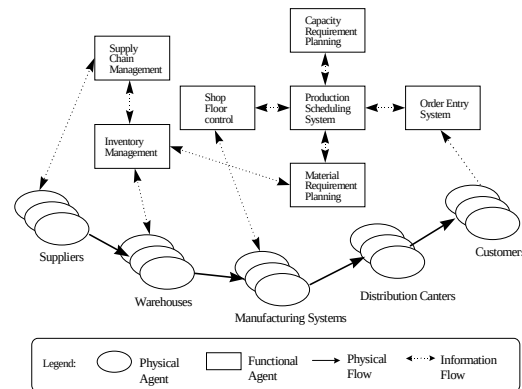


Figure 3. The MAIS utilizing physical and logical agent representation for the OFP

The Swarm simulation platform is suitable for modeling supply chain networks based on the following analysis on their properties (summarized in Table 2). A supply chain network is composed of several autonomous or semi-autonomous business entities which can be viewed as a swarm of agents. Each business entity has its capability and capacity and can be assigned to, or take certain types of tasks, according to its organizational

roles in the SCN. These capability, capacity, and organizational roles can be modeled by individual-based models. The SCN can be decomposed according to the multiple layer abstraction, which can be modeled by Swarm's nested inherent hierarchy property. The material flow is controlled by business entities within the SCN. These business entities can be represented as physical agents, while logical agents are used for controlling actions for material movement. These operational and control actions can be simulated by Swarm's hybrid scheduling capability (i.e., discrete event and time-stepped scheduling). SCN visibility depends on the boundary of information sharing which is an analogy to the boundaries of message passing in Swarm. In a SCN, the processes of each individual entity contribute to the global performance of the SCN, while in Swarm, the combination of individual behaviors determine the collective performance of the whole group. We enhanced the Swarm functions and customized it to model the OFP in SCNs.

Table 2. The mapping of properties between SCNs and Swarm

Supply Chain Networks	Swarm Simulation System
Composition of autonomous and semi-autonomous business entities	A swarm of agents with individual based modeling
Business entities act different organizational roles.	Agents are constructed with internal state variables and action functions.
Multiple layer abstraction	Nested inherent hierarchy
Information flow between business entities	Message passing between agents
Material flow during procurement, manufacturing, and distribution activities	Discrete event simulation and time-stepped scheduling to trigger agent actions
In a SCN, the processes of individual entities contribute to the global SCN performance.	In Swarm, the combination of individual behaviors determines the collective group behavior.
The visibility of a SCN is determined by the information boundary.	The boundaries of message passing determine the visibility.

Figure 4 shows the resulting system developed to model the OFP in SCNs based on the Swarm. The topmost Swarm, *OFP Batch Swarm*, is designed to control the whole simulation. It creates two swarms, the *OFP Model Swarm* and the *Statistics Swarm*, creates actions, and then activates the simulation process. The *OFP Model Swarm* is composed of an array of *SCN Entities* created while building objects. The SCN

configuration with each entity's properties and product information are fed in during the entities creation. The OFP model actions are composed of each SCN entity's action, and then activate them when the OFP Model Swarm is activated. A SCN entity is composed of several agents, such as an order management agent, an inventory management agent, a SCN management agent, and so forth. In addition to having an order management agent, an inventory management agent, and a SCN management agent, an entity with manufacturing capability includes a production planning agent, a capacity planning agent, a material planning agent, a shopfloor control agent, and manufacturing systems. A SCN Entity Swarm holds entity level information such as suppliers, customers, order transfer delay time, and product delivery time, which are accessible by internal agents and other entities. The encapsulated agents perform certain functions in enabling the movement of information and material within the entity and between entities. The *OFP Statistic Swarm* is used to compute the statistics data gathered through the simulation for analysis purposes.

The functions of agents are described as follows:

- *Order management agent* -- It processes orders coming from outside entities or other agents inside the entity. Tasks include estimating due dates, checking inventory availability, and forwarding orders to upstream suppliers if needed. It obtains orders from the SCN Swarm, and queries inventory availability through the inventory management agent. If the entity has manufacturing capability, the estimation of due dates also queries product availability through the production planning agent and the shop floor control agent.
- *Inventory management agent* -- It handles the inventory related tasks, such as keeping track of inventory records, determining the reordering point, and responding to the inventory availability queries from other agents, such as order management, material planning, and shop floor control agents.
- *Production planning agent* -- It takes the orders from the order management agent, and then, according to the product due dates, capacity and material availability, generates build plans under different planning horizons. The build plan will be used by the material planning and the shop floor control agents to execute the production process.
- *Capacity planning agent* -- It keeps track of the capacity consumption of manufacturing systems, and provides capacity availability information for the production planning agent in production

planning.

- *Material planning agent* -- It has two main roles: one to plan the required materials for the manufacturing process, including the allocation of parts from inventory, or the purchase of materials from suppliers; the other is to provide material availability information for the production planning agent to generate build plans.
- *Shopfloor control agent* -- It dispatches parts from inventory to the manufacturing systems according to the work order derived from build plans. It also provides capacity utilization information for other agents.
- *Manufacturing agent* -- It executes the manufacturing processes by inputting the components and utilizing capacity.
- *SCN management agent* -- It chooses suppliers for the entities in acquiring components for manufacturing or end products for distribution.

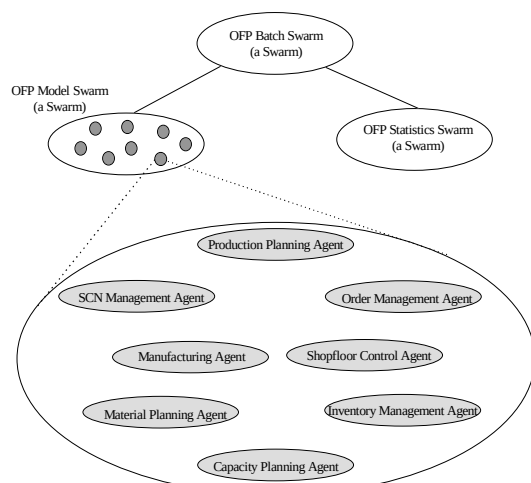


Figure 4. The implementation of Swarm in modeling supply chain networks

The following scenario describes the interactions among these agents, and Figure 5 summarizes them. For example, an entity *ScnESwarm A* receives an order from its customer *ScnESwarm C*. The order flows to the order management agent (*OrdM*). According to the customer lead times, the inventory availability information (from *InvM*), the production plan (from *PrdP*), and the manufacturing capacity (from *CapP*), the order management agent assigns a due date to the order. If the products are in stock, the order is fulfilled by shipping the products from inventory. If the products are in receiving, the due date is set according to the delivery delay of the products. For an entity with manufacturing capability, the order is forwarded to the

production planning agent (*PrdP*) where the schedule for making the products is planned. The capacity planning agent (*CapP*) and the material planning agent (*MatP*) are partner agents in generating achievable build plans. The material planning agent obtains build plans from the production planning agent to allocate materials for manufacturing. It also contributes information about material availability to production planning for scheduling. The capacity planning agent (*capP*) plans the capacity by taking the build plan from *PrdP* and sends capacity usage information to *PrdP* for scheduling the build plan. The SCN management agent (*ScnM*) takes the order information to choose suppliers in allocating material sources. The outgoing orders are transferred through its SCN Entity (*ScnESwarm A*) to be transferred to other entities (i.e. *ScnESwarm B*). The above describes the *information flow* within an entity.

If the entity is a distribution center or a retailer without manufacturing capability the ordered products are delivered from suppliers as end products to ship to its customers. For an entity with manufacturing capability, the ordered end products are supplied from the shop floor (*ManuS*) to its customers. The input materials are components for the end products. This represents the *material flow* within an entity. The interaction of these agents enables the flows of materials and information within an entity, and through SCN Entity Swarm (*ScnESwarm*), the information and material flow across the supply chain network.

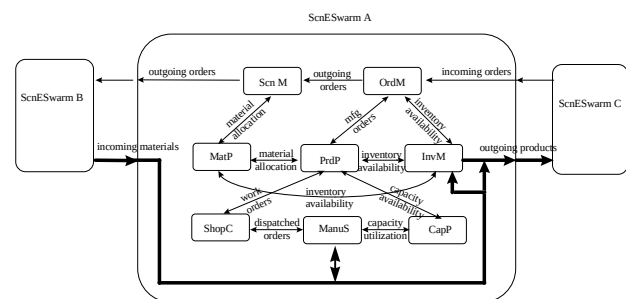


Figure 5. The scenario of OFP within an entity

6. Evaluating OFP Improvement Strategies

In order to achieve OFP agility, the efforts should be spent on problems involved with the issues in improving the OFP in SCNs. The proposed strategies for improving the OFP in SCNs are based on the dimensions of managing supply chain networks, such as the operations, structures, information infrastructure of the supply chain network, and the other supporting business processes. Experiments are performed by modeling the

OFP in SCNs based on the Swarm simulation system. The strategies we evaluated include

- (1) *Coordinating demand management policies* -- The strategy of coordinating demand management policies among entities across the network is aimed at satisfying customer demand with low inventory costs. Because inconsistent demand management policies will cause material shortages or overflow, coordination of demand management policies is key in resolving this problem. In the design of the simulation platform, four demand management policies are embedded within each entity, that is, *Make-To-Order* (MTO), *Assembly-To-Order* (ATO), *Make-To-Stock* (MTS), and *Build-To-Forecast* (BTF). For entities with manufacturing capability, these policies trigger their manufacturing processes. For entities without manufacturing capability, these policies describe different events for re-ordering products. Depending on the number of tiers in the SCN being modeled, various demand management policies are applied to different tiers, and OFP performance is evaluated using the performance measures from above. The effects of coordinating demand management policies can be simulated through various combinations of different policies across the network.
- (2) *Information sharing strategies* -- The strategy of sharing information between business entities considers three issues: *the information contents*, *the depth of information penetration* (the number of tiers for which information is accessible), and *the information acquisition direction* (upward or downward sharing). In the design of the simulation platform, the information acquired by downstream entities are mainly material and capacity availability information from their suppliers. The information acquired by an upstream entity is information about customer demand and orders. The depth of information penetration can be specified in various degrees, e.g., isolated, upward one tier, upward two tiers, downward one tier, downward two tiers, and so forth. The obtained capacity and material information from suppliers is used to estimate the due dates of incoming orders, which are the basis for generating build plans or re-ordering schedules. The obtained customer demand information is used to estimate the demand for the next period, so that the production or re-ordering schedules can adapt to external demand.
- (3) *Synchronizing material and capacity availability* -

This strategy is aimed at enabling the production planner to generate more achievable build plans by considering both material and capacity availability. A build plan cannot be executed if either capacity or material is unavailable. The material and capacity information shared by the production planner can improve the accuracy of build plans. The effects of the synchronization can be evaluated by the Swarm simulation platform.

- (4) *Dynamic material allocation* -- This strategy is aimed at maintaining a SCN's robustness while facing uncertainties, such as demand and supply fluctuations, and production process interruption. The application of this strategy can be evaluated using the Swarm simulation platform.
- (5) *The combination of various strategies* -- This approach is to combine the usage of the above strategies. The evaluation aims at finding the joint effects of different strategies.

These strategies were evaluated under a generic and three specific supply chain networks with various demand management policy combinations, different lead times, and uncertainties. The performance of the order fulfillment process is measured by *the order cycle time*, *the order fulfillment rate*, *the inventory cost*, and *tardiness on committed due dates and customer expected lead times*. The efforts are spent to understand interactions between various demand management policy combinations, to understand the main improvements from various OFP improvement strategies, and to obtain the trends of OFP improvement using different strategies under different circumstances. We describe the evaluation results as follows.

The OFP performance using different combinations of demand management policies can be generalized as the following patterns for applying combinations of policies to SCNs:

- *The cycle time is a tradeoff with inventory cost.* Entities applying MTS or ATO result in shorter order fulfillment cycle times, however, their inventory costs are higher than other policies. Due to the shorter cycle time, they create less order fulfillment tardiness problem both measured in committed due dates and customer expected lead time.
- The better policy combinations are those that use a MTO policy at downstream tiers, and use MTS or ATO policies at upstream tiers. The distinct results are shown in the extremely opposite combinations, BTF-MTS-ATO-MTO vs. MTO-ATO-MTS-BTF.

- The variation of lead times among very short, short, and long does not change the relative performance of different policy combinations. Therefore, the coordination of different policies is applicable to various lead times.
- The variation of demand patterns between even and uneven does not change the relative performance of different policy combinations.
- One unexpected result is that the BTF policy does not contribute significant improvement. The reason may be twofold: one reason could be that the forecasting method does not accurately forecast demand from previous demand patterns; the other could be the period of forecasting is too short for entities to prepare, so that it works more like a MTO policy.

The trends of information sharing strategies on improving OFP performance are described as follows: The main effects of upward information acquisition include: (1) increasing the order fulfillment rate for those policy combinations having low throughput at downstream tiers, (2) significantly reducing inventory cost at an assembly tier for those entities using make-to-stock or assembly-to-order policies, and (3) significantly reducing tardiness at upstream tiers and downstream tiers. So that the main effects of upward information sharing are on reducing the inventory cost and tardiness, and increasing the order fulfillment rate. The main effects of downward information acquisition include: (1) trading the cycle time with the inventory cost at those policy combinations having the short order cycle time, and (2) significantly reducing the inventory cost at an assembly tier whose entities use an assembly-to-order demand management policy. These results support that downward information sharing substitutes information for inventory. The results of bi-directional information sharing show that the order cycle time is mainly affected by the downward information sharing strategy. This support that the degree of OFP improvement is more heavily affected by the accuracy of demand information from customers than that of the material availability information from suppliers. However, for the other performance measures, both upward and downward information sharing strategies equally contribute to the performance improvement, which results in the greater improvement than when only either one is applied. This enhancement supports that the degree of OFP improvement by a bi-directional information sharing is more significantly than that by a single direction. The marginal improvements of multiple tiers of information sharing against single tier information sharing are mainly on inventory cost reduction and due date commitment. For other measures, there are no significant

improvements. Therefore, the marginal improvement of using multiple tier information sharing is evaluated by how critical the inventory and tardiness reduction compared with the information cost increased by the multiple tier information sharing.

Based on the results we found that by synchronizing the material and capacity availability, the improvement is not only on one single entity, but also improve the OFP across a SCN. The main effects of this strategy are shown on the inventory reduction for those manufacturing entities, and on the tardiness reduction for the downstream customers. In general, the strategy for synchronizing capacity and material availability is an effective strategy in meeting customer demand with less inventory and tardiness.

Generally, from experiments on the dynamic material allocation strategy, we can conclude the following common trends of its effects: (1) the dynamic material allocation strategy is an effective approach to reduce OFP vulnerability under supply uncertainties, and to improve OFP robustness under demand uncertainties, and (2) the effectiveness of dynamic material allocation strategy is highly influenced by the information transfer efficiency. With efficient information exchange, the order cycle time and order fulfillment rate are improved with less tardiness compared with material allocation from stable suppliers. However, with inefficient information exchange, the OFP cycle time and tardiness may increase although the fulfillment rate can increase on those good policy combinations.

The results of OFP improvement by combining the synchronization with information sharing strategies show the synergy of the two strategies. By combining the demand or supply information from external entities in production planning, the order fulfillment rate at downstream entities, such as distribution centers and retailers, increases because the manufacturing entities can deliver goods in a timely manner. Because of the limitation of manufacturing capacity, the fulfillment rate is reduced while the demand exceed the capacity, which lengthens the order fulfillment cycle time. Inventory costs are reduced across tiers except at the assembly tier, the increase of the fulfillment rate increases the inventory, but not significantly. Under demand uncertainties, the joint strategy also shows its strengths to improve the OFP, where OFP robustness is shown using this strategy. The analysis of the joint effects of dynamic material allocation and synchronization of capacity and material availability shows that such strategy significantly reduces cycle times, inventory costs, and tardiness.

The findings from experiments are summarized as follows:

1. **Information can substitute for inventory.** The information sharing strategies by querying supply information from suppliers and demand information from customers greatly improves the OFP in terms of reducing inventory costs, order cycle time and tardiness, and increasing the order fulfillment rate. Demand information is more critical than supply information because an accurate demand estimation facilitates the usage of supply information. The bi-directional information sharing by sharing supply and demand information can further improve the OFP than uni-directional information sharing by sharing either supply or demand information. The deeper the information is penetrated the better the OFP is improved, especially at inventory and tardiness reductions. The information sharing strategies can improve the OFP under various lead times and uncertainties. In general, information substitutes for inventory in eliminating the variability of a SCN and improving the OFP's robustness.
2. **Information infrastructure is critical for OFP improvement.** The coordination among agents within an entity or among entities in a SCN relies on efficient information transfer. For example, the OFP is not improved by using the dynamic material allocation strategy without efficient information transmission. Because this strategy requires several iterations of information exchange between entities, it actually delays the OFP when the information transmission speed is below or equal to the speed of order or material transportation.
3. **Different strategies have their marginal improvement in different SCNs with different policies.** From the experiments on evaluating various strategies in a generic supply chain network, we found that OFP improvement varies under different demand management policy combinations using various strategies. However, there are some trends can be generalized to understand the effects of OFP improvement strategies on different SCNs:
 - A. Supply information is important for an entity in a SCN when end products are assembled by many components from different suppliers. Therefore, an upward information sharing strategy is applicable.
 - B. Demand information is critical for an entity in SCN when it faces a constantly changing markets and high customization pressure. Therefore, a downward information sharing strategy is applicable.
 - C. The synchronization strategy to coordinate material arrival and capacity allocation is

beneficial when products are made by many constituent components.

- D. Demand management can use the assembly-to-order policy when information sharing or synchronization strategies are utilized.

Through the modeling and evaluation efforts, an enterprise can envision its OFP performance under different strategies within various business environments. The MAIS approach is a viable method to model the OFP in SCNs, and the Swarm simulation platform is a useful implementation tool. Therefore, the proposed strategies can be accurately evaluated before the OFP is redesigned, and moreover, better strategies can be derived by generalizing the evaluation results. Such strengths make the MAIS approach applicable to other business process reengineering because various business processes possess several common properties with the OFP, such as cross organizational boundaries, task interdependence, and various degrees of autonomy for business entities.

7. Conclusions

In this article, we propose the multi-agent information system (MAIS) approach to model the order fulfillment process (OFP) in supply chain networks (SCNs), and evaluate strategies for improving OFP performance. Through the modeling and implementation of the MAIS using Swarm, we demonstrate that the MAIS approach is a viable method to model the OFP in SCNs, and identify and evaluate some strategies for improving OFP performance. The improvement of the OFP by various strategies with various measures provides insights to manage the supply chains for fulfilling orders. Moreover, the MAIS approach is also suitable for reengineering other business processes because business processes share several common characteristics.

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